

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.4: Electrostatics — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.4: Electrostatics
Driving Question	DRIVING QUESTION: How do invisible electric charges build up, exert forces across empty space, and store energy — and how can we harness these abilities to protect communities and power technology in Kenya?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is Happening Before the Lightning Strikes? (Introducing Charge)	Students observed: (1) a balloon rubbed on hair or wool attracting small paper pieces and repelling another similarly charged balloon; (2) gold leaves of an electroscope diverging when a charged rod was brought near; (3) a charged balloon failing to attract paper through a wooden board but showing an effect through thin cardboard; (4) a video of the Ol Kalou lightning strike reconstruction, a leaf electroscope in action, and a Van de Graaff generator causing hair to stand on end at a Nairobi science fair.	Students learned that: (1) all matter is made of atoms containing positive protons and negative electrons; (2) objects become electrically charged when electrons are transferred by friction (triboelectric effect) — the object gaining electrons becomes negative, the one losing electrons becomes positive; (3) like charges repel and unlike charges attract; (4) conductors (such as iron sheet) allow electrons to move freely, while insulators (such as wood) do not — this is why the iron-roofed stall at Ol Kalou was struck while the wooden stall was untouched.	Students began to explain: (1) why the traders' arm hairs stood on end before the strike — charge built up in the cloud induced charge separation in nearby objects, including the hairs, which repelled each other; (2) why the transistor radio crackled — rapidly changing electric fields from charge build-up interfered with the radio signal; (3) why the iron roof was struck — as a conductor, it allowed charge to concentrate and provided a low-resistance path. Students could NOT yet fully explain how charge built up in the cloud, how the lightning conductor directed the strike safely into the ground, or how forces act across empty space — these puzzles are posted on the DQB and will be addressed in Lessons 2–9.
Lesson 2 Two Types of Charge, the Law of Charges, and Conductors vs. Insulators	Learners observed that two balloons both rubbed with wool repelled each other, while a balloon rubbed with wool and one rubbed with nylon attracted each other. A charged plastic comb attracted torn paper scraps ('mapande ya karatasi'). Iron wire and copper wire from a jua kali workshop lit a	There are two types of electric charge — positive and negative. Charging by friction transfers electrons: the object gaining electrons becomes negative; the object losing electrons becomes positive. Like charges repel; unlike charges attract (law of charges). Conductors (e.g., iron, copper)	The iron roof at Ol Kalou is a conductor — its free electrons allowed the enormous charge built up in the storm cloud to flow rapidly through the roof, causing the scorch and melting the iron sheet. The wooden stall is an insulator — electrons cannot move freely through wood, so the charge

	circuit-tester bulb; sisal twine, dried maize husk, and a rubber slipper sole did not.	allow electrons to move freely; insulators (e.g., wood, sisal, rubber) resist electron flow. This explains why the iron roof at Ol Kalou was a pathway for the lightning discharge while the wooden stall was not.	had no pathway through it and the stall was undamaged. The traders' arm hairs stood on end because the large charge nearby exerted an attractive force on the opposite charges in their hair, pulling the hairs toward the cloud — a force-at-a-distance effect that Lesson 3 will explore further.
Lesson 3 How Strong Is the Push or Pull Between Charges? (Coulomb's Law)	We observed that two charged balloons repel each other and that the repulsion weakens noticeably as they are moved further apart. We also analysed data from a torsion-balance experiment and plotted a straight-line graph of F versus $1/r^2$.	We learned that the electrostatic force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them: $F = kQ_1Q_2/r^2$, where k is approximately $9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.	This explains why the lightning strike at Ol Kalou became inevitable as the storm cloud descended closer to the iron roof: as r decreased, the force increased as $1/r^2$, rapidly overcoming the insulating strength of air and triggering the discharge.
Lesson 4 Charging by Contact, Induction, and Earthing — How Did the Iron Roof Gain Charge Without Touching the Cloud?	Learners observed that bringing a charged PVC pipe near (without touching) the homemade aluminium-foil electroscope caused the foil leaves to diverge; that touching the ball with a finger while the charged rod was held nearby and then removing the finger BEFORE the rod left the leaves permanently diverged even after the rod was removed; and that the tanker drag chain on Mombasa Road provides a conducting path between the tanker body and the ground.	Learners learned that charging by induction redistributes charge within a conductor without any physical contact between the charging object and the conductor; that earthing (grounding) removes excess charge by providing a conducting path to the Earth; that the sign of charge acquired by induction is opposite to the sign of the inducing charge; and that insulators (like the wooden stall) cannot be charged by induction because they have no free electrons to redistribute.	Learners explained that the iron roof at Ol Kalou Market became positively charged on its upper surface through induction — the negatively charged storm cloud repelled free electrons in the iron downward, leaving a positive charge on the surface closest to the cloud — without any contact between the cloud and the roof. This charge separation created the conditions for the lightning strike. A lightning conductor prevents damage by keeping the building connected to the Earth, so any induced charge drains away harmlessly rather than building to a discharge.
Lesson 5 How Strong Is the Push or Pull Between Charges? — Coulomb's Law	Students observed (via PhET simulation or torsion-balance demonstration) that the electrostatic force between two charged objects decreases as separation increases, and increases as charge magnitude increases. They recorded quantitative data and plotted F -vs- r and F -vs- q graphs, identifying a curved (inverse-square) relationship for distance and a linear relationship for charge.	Students learned Coulomb's Law ($F = kq_1q_2/r^2$, $k \approx 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$) and applied it to calculate electrostatic forces in Kenyan contexts, including the force between the storm cloud and the iron roof at Ol Kalou Market. They discovered that doubling the distance reduces the force to one quarter, explaining why taller conductors with smaller separation from the cloud are preferred discharge paths.	Using Coulomb's Law, students explained why the iron-roofed market stall — not the wooden stall beside it — was struck by lightning: the iron roof had a significant induced charge ($q_2 \neq 0$) and was the closest large conductor, maximising $F = kq_1q_2/r^2$. They also explained why the lightning conductor on the district offices 30 m away could intercept the strike — its height minimised r and its conductive tip concentrated induced charge, producing a larger attractive force than either stall.
Lesson 6 What Is a Capacitor and How Does It Store Charge? (Capacitance)	We constructed a parallel-plate capacitor from aluminium foil and a plastic bag, charged it from a 9 V battery, and discharged it through an LED. We observed	A capacitor stores electric charge on two conducting plates separated by an insulating dielectric material. Capacitance (C) is defined as the charge stored per unit	Before the lightning strike at Ol Kalou, charge separation between the storm cloud and the iron roof created a situation identical to a charged capacitor — two

and the Capacitor)	that greater plate area produced a brighter or longer LED flash, and that greater plate separation produced a dimmer or shorter flash. We also observed that the LED lit up after the battery was disconnected, proving energy had been stored in the capacitor.	potential difference: $C = Q/V$, measured in farads (F). Capacitance increases with larger plate area and decreases with greater plate separation. The dielectric material also affects how much charge can be stored. The storm cloud above Ol Kalou acted as a natural parallel-plate capacitor, with the cloud base and the iron roof as its two plates and the air as its dielectric.	conductors at different potentials separated by a dielectric (air). As the potential difference grew, so did the stored charge ($Q = CV$). When the electric field in the air gap became strong enough to ionise the air, the capacitor discharged catastrophically as a lightning bolt. The energy stored in this natural capacitor — which we will calculate in Lesson 8 — was large enough to melt iron sheet and generate the static that crackled in the nearby radio.
Lesson 7 How Does Charge Store Energy — and How Can That Energy Be Released or Harnessed?	Students observed: (1) a pith ball deflecting progressively as a Van de Graaff generator charged — then snapping back to vertical upon discharge; (2) a homemade aluminium-foil parallel-plate capacitor storing charge at three different voltages (1.5 V, 3 V, 4.5 V) and releasing it to flash an LED or deflect a galvanometer; (3) a class Q-V graph forming a straight line through the origin, with a shaded triangular area representing stored energy.	Students learned that: (1) work done separating charges is stored as electric potential energy; (2) electric potential (voltage) equals work done per unit charge ($V = W/q$); (3) capacitance ($C = Q/V$) measures how much charge a device stores per volt; (4) energy stored in a capacitor equals the area under the Q-V graph ($E = \frac{1}{2}CV^2$); (5) the cloud-to-ground gap in the Ol Kalou storm acted as a natural parallel-plate capacitor storing up to 1.5×10^9 J before catastrophic discharge.	Students explained that: (1) the energy scorch-marking the iron-sheet roof at Ol Kalou had been stored invisibly in the electric field between the charged storm cloud and the ground — exactly as energy is stored in the field between the foil plates of their homemade capacitor; (2) the lightning conductor on the district offices provided a controlled discharge pathway, safely routing stored energy to ground rather than through the iron roof; (3) engineers exploit the same charge-storage principle in camera flash units, mobile phone chargers, and surge protectors — devices that charge slowly and release energy rapidly or safely absorb excess energy.
Lesson 8 How Much Energy Can a Capacitor Store? (Energy Stored in a Capacitor)	We explained that the energy stored in a capacitor is the work done by the battery in separating charges onto the plates against the growing electric field. Because voltage increases as charge builds up, the work done per extra unit of charge also increases, so the total energy is the triangular area under the Q-V graph rather than a simple rectangle. This is captured exactly by $E = \text{half } CV \text{ squared}$.	We learned that the energy stored in a capacitor equals half times C times V squared, that voltage has a squared effect on stored energy (doubling the voltage quadruples the energy), and that this formula applies equally to small engineered capacitors and to the enormous natural capacitor formed between the storm cloud and the ground above Ol Kalou.	The energy stored in a capacitor is not simply charge times voltage — it is half that, because the voltage starts at zero and grows as charge accumulates, so the average voltage across the charging process is $V/2$. This is why doubling the voltage on a capacitor quadruples the energy it stores: the formula $E = \text{half } CV \text{ squared}$ has voltage squared, not linear. The same physics explains the enormous energy released in a lightning strike above Ol Kalou — the natural capacitor formed between the storm cloud and the ground charges up to millions of volts, and even though its capacitance is modest, the V squared term makes the stored energy catastrophic when it discharges.

Lesson 9 How Do We Use Electrostatics to Solve Real Problems? (Applications and Unit Synthesis)	We observed five student group presentations on electrostatic applications operating in Kenya — lightning conductors in Nyandarua, electrostatic precipitators at Athi River, photocopiers in Kisumu, spray-painting booths on Mombasa Road, and inkjet printers — and we completed the Driving Question Board by placing green dots on resolved questions from Lesson 1.	We learned that all five applications rely on the same fundamental processes — charge separation or induction, electric field concentration or direction, controlled potential difference, and energy transfer — that we used to explain the OI Kalou lightning strike, confirming that electrostatics is a single coherent framework operating at every scale from a printer droplet to a storm cloud.	We explained the complete OI Kalou Market event in a nine-step narrative linking charge build-up by friction in the cloud, induction on the iron roof, Coulomb force intensification, field concentration at the conductor tip, ionisation of air at breakdown potential difference, rapid capacitor-like discharge releasing energy equal to half CV squared, selective damage to the conducting iron roof but not the insulating wooden stall, and the safe earthing pathway provided by the lightning conductor — and we connected this explanation to at least one engineered technology that harnesses the same physics productively for Kenyan communities.
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